TITLE: COMPARISON OF PLANT CELLS – 30 min

Time required
30 minutes

Equipment

- 1 × Optical microscope
- 2 × Microscope slides
- 2 × Cover slips
- 1 × Tweezers
- 1 × Absorbent paper sheet (not included)
- 1 × Dropper

EXPERIMENT IMAGES

STEP 1: Slice of white onion cut
STEP 2: Epidermis sample placed on a microscope slide
STEP 3: Staining of the epidermis sample with dye solution
STEP 4: Observation of the sample with the second cover slip

(Magnification: 40x)

Objective of Experiment No. 1

The objective of this experiment is to compare the structure of two different plant cells observed under a microscope. In particular, we will examine the structure of cells from **white onion** and from **Phalaenopsis orchid**.

Description of the experiment

- Cut a slice of white onion with a kitchen knife and take a scale (STEP 1).
- Take a clean slide and place a drop of distilled water in the center using the dropper. Then, using tweezers, carefully remove the epidermis of the onion scale (a thin, transparent surface layer). Arrange the sample properly to hydrate it (STEP 2).
- Using tweezers, spread the epidermis sample well over the drop of water. Then pour 50 ml of distilled water into a beaker and add 5 drops of methylene blue to obtain an intense blue aqueous solution. Using a dropper, place two drops of this solution on the sample as a contrast agent (STEP 3).

- Carefully place a second cover slip over the prepared sample, making sure not to create air bubbles that would prevent clear observation under the microscope. If necessary, dry the excess methylene blue solution that leaks out, then place the sample under the microscope (STEP 4).
- Focus the microscope and carefully examine the slide first at low magnification (eyepiece 10x, objective 20x) and then at higher magnification (eyepiece 10x, objective 40x).
- Record the observations of the onion sample and repeat the same procedure with a **Phalaenopsis orchid leaf**.

Final Observation

From this experiment, we can observe that onion cells are more elongated and flattened, whereas the cells of the *Phalaenopsis* orchid leaf are polygonal. In the plant cells of the *Phalaenopsis* orchid, chloroplasts are present—organelles located near the cell membrane and responsible for photosynthesis—as well as stomata, which are not observed in the cells of the white onion

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Questions for the teacher

1. Why are chloroplasts clearly visible in orchid cells, unlike in onion cells? What would happen if orchid cells lacked chloroplasts?
2. What would we observe if we analyzed an epidermis sample taken from a celery stalk? Would celery cells show a strong tendency toward hydration or not?
